Arman Simonyan

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Professional & Technical Skills

- **PhD candidate (2022–2026)** in AI-driven drug discovery for **G protein-coupled receptors (GPCRs)** at University of Copenhagen; h-index 6, 7 peer-reviewed publications, with manuscript applying AlphaFold-based peptide design to neurological targets in preparation — directly relevant to Lundbeck's preclinical neuroscience portfolio.
- **7+ years** building bioinformatics data pipelines in **Python** and **R**; cloud infrastructure (**AWS** S3/EC2/CDK), containerization (**Docker**), and workflow orchestration (**Nextflow**, **Snakemake**) across HPC and cloud environments.
- Extensive experience processing diverse preclinical scientific data types: bulk, single-cell and spatial **RNA-seq**; whole-exome/genome sequencing; **GWAS**; time-series electrophysiology; fluorescent and confocal imaging; and large video files.
- Applied **deep learning** (GNNs, attention, VAEs, protein language models) to drug target identification and ligand design; **Danish Data Science Academy PhD Fellow** (2023); two **International Biology Olympiad Bronze Medals** (2014 Indonesia, 2015 Denmark).
- Cross-functional collaborator across computational and wet-lab teams at **University of Copenhagen**, **Rigshospitalet** (Center for Genomic Medicine), and **Harvard Medical School** (Dystonia Neuroscience Group); Copenhagen-based, immediately available.

Scientific Career

University of Copenhagen, *DK*

2022-26: **Ph.D. in Pharmaceutical Sciences** (Gloriam Group ILF)
Thesis: AI-powered drug discovery for G-protein coupled receptors
Key competences: Protein/peptide engineering (RFdiffusion, AF-backpropagation, ProteinMPNN, ESM-IF), Protein-binder co-folding (AF2/3, Bolz2, ESM), Homology Modeling, Docking, Molecular Dynamics. Cloud-based scientific data pipelines (AWS), workflow orchestration (Nextflow/Snakemake), containerized workloads (Docker), multi-omics and time-series data integration.

2020-22: **M.Sc. in Bioinformatics**
Thesis: *Automated pharmacogenomic decision support for inclusion in Phase 1 clinical trials*
Key competences: Biological Sequence Analysis (DNA, RNA, spatial-seq, MSA, Assembly, Variant Calling, GWAS, QTLA), Functional Analysis, Protein Language Models.

Rigshospitalet, *DK*

2020-22: **Bioinformatics research assistant** (Center for Genomic Medicine)
Projects: - Identification of driver-mutations in oral cavity squamous cell carcinoma
- Characterization of nonsense-mediated decay in cancers

Harvard Medical School, *USA*

06-08/2019: **Researcher-in-residence** (Dystonia Group, MEEI/HMS)
Project: *Machine learning for identification of genetic variants with neural correlates in spasmodic dysphonia*

Institute of Molecular Biology *NAS, Armenia*

2018-20: **Researcher (**Bioinformatics Group**)**
Projects: - Transcriptome-based drug repositioning platform development based on biological pathway curation
- Computational approaches for investigating the role of telomeres in molecular mechanisms of disease progression

2017-18: **Research Assistant (**Human Genomics lab**)**
Projects: Characterization of genetic markers for schizophrenia in Armenian population
Key competences: DNA extraction & purification, PCR, Gel electrophoresis

Orbeli Institute of Physiology *NAS, Armenia*
9-12/2017: **Researcher-in-residence** *(Tissue Engineering Lab)*
Key competences*: Bioreactor engineering, tissue harvesting, Fluorescent & Confocal imaging, Mass spectrometry*

American University of Armenia
2016-20: **B.Sc. in Computer Science** & Math Modeling
Thesis: Analysis of transcriptome and survival patterns in BRCA1- and BRCA2-positive breast and ovarian cancers

Skills

DL Architectures: Graph Neural Networks, Attention, Self-Organizing Maps, Variational Autoencoders.
Programming: Python, R, Bash, Java, C++, git, HPC; AWS (S3, EC2, CDK), Docker, Nextflow, Snakemake.
Scientific Data: Omics (bulk/single-cell/spatial RNA-seq, WES/WGS, GWAS), time-series electrophysiology, imaging (fluorescent, confocal), large-file video data.
Spoken Languages: Armenian (native), English (proficient), Russian (proficient), Danish (conversational, B1).

Selected Publications

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Simonyan, A., Kulkarni, Y., Buzón, A.R., Zhang, U., Holst, B., Boomsma, W., Gloriam, D.E.
Human-, Rosetta- versus AlphaFold-based design of Ghrelin receptor ligands
Manuscript in preparation

Probst, V., **Simonyan, A.**, Pacheco, F., Guo, Y., Nielsen, F.C., Bagger, F.O.
Benchmarking full-length transcript single cell mRNA sequencing protocols.
BMC Genomics, 2022.

Nersisyan, L., **Simonyan, A.**, Binder, H., Arakelyan, A.
Telomere maintenance pathway activity analysis enables tissue- and gene-level inferences.
Front. Genet., 2021.

Arakelyan, A., Melkonyan, A., Hakobyan, S., Boyarskih, U., **Simonyan, A.**, Nersisyan, L., Nikoghosyan, M., Filipenko, M., Binder, H.
Transcriptome Patterns of BRCA1- and BRCA2- Mutated Breast and Ovarian Cancers.
Int. J. Mol. Sci., 2021.

Simonyan, A. (2020).
Quantitative Trait Locus Analysis in Spasmodic Dysphonia. In: Darbinyan A. (eds).
Proceedings of the XIV Annual International Conf. of Russian-Armenian University. Yerevan, Armenia: Russian-Armenian University. 2020.

Simonyan, A., Sarvazyan, N. (2020).
Bioreactors. In: Sarvazyan N. (eds).
Tissue Engineering. Learning Materials in Biosciences. Cham, Switzerland: Springer International Publishing.

Arakelyan, A., Nersisyan, L, Nikoghosyan, M, Hakobyan, S, **Simonyan, A.**, Hopp, L., Loffler-Wirth, H, Binder, H.
Transcriptome-guided drug repositioning.
Pharmaceutics, 2019.

Simonyan, A., Stepanyan, A.
Analysis of transcriptome changes in response to heavy metals using self-organizing maps.
Katchar Scientific Periodical, 2019.

Conference Talks

2026: Human-, Rosetta- versus AlphaFold-based design of Ghrelin receptor ligands
2nd GPCR FORUM virtual conference

2025: New ML-Driven Approaches for Peptide Hormone and Drug Discovery
76th Mosbacher Kolloquium "AI-driven (r)evolution in structural biology & protein design" (Germany)

2020: Analysis of transcriptome changes in response to heavy metals using self-organizing maps
Rīga Stradiņš University Intl. Student Conf. 2020 (Latvia)

2019: Genome-wide Association Study in Laryngeal Dystonia
XIV Annual Intl. Conf. of Russian-Armenian University (Armenia)

2019: Tissue engineered constructs and 3D bioprinting
XXVI Intl. Student Conf. "Lomonosov 2019" (Russia)

2018: Comparative study of liver bioengineering protocols
Intl. Scientific Conf. "Kliment's Days" 2018 (Bulgaria)

Grants and Awards

2023: Danish Data Science Academy PhD Fellowship (*Denmark*)

2020-22: Danish Governmental Scholarship for Non-EU students

2022: Best Presentation Award in RSU int. conference (*Latvia*)

2019: State Award for The Best Student in IT (*Armenia*)

2019: AGBU Summer Research Scholarship (*USA*)

2019: Best Talk in "Lomonosov 2019" int. conference (*Russia*)

2016: Presidential Award for The Best Student in IT (*Armenia*)

2016-20: Academic Excellence Scholarship of American University of Armenia

2015: **Bronze medal** in 26th International Biology Olympiad (*Denmark*)

2014: **Bronze medal** in 25th International Biology Olympiad (*Indonesia*)

References available upon request